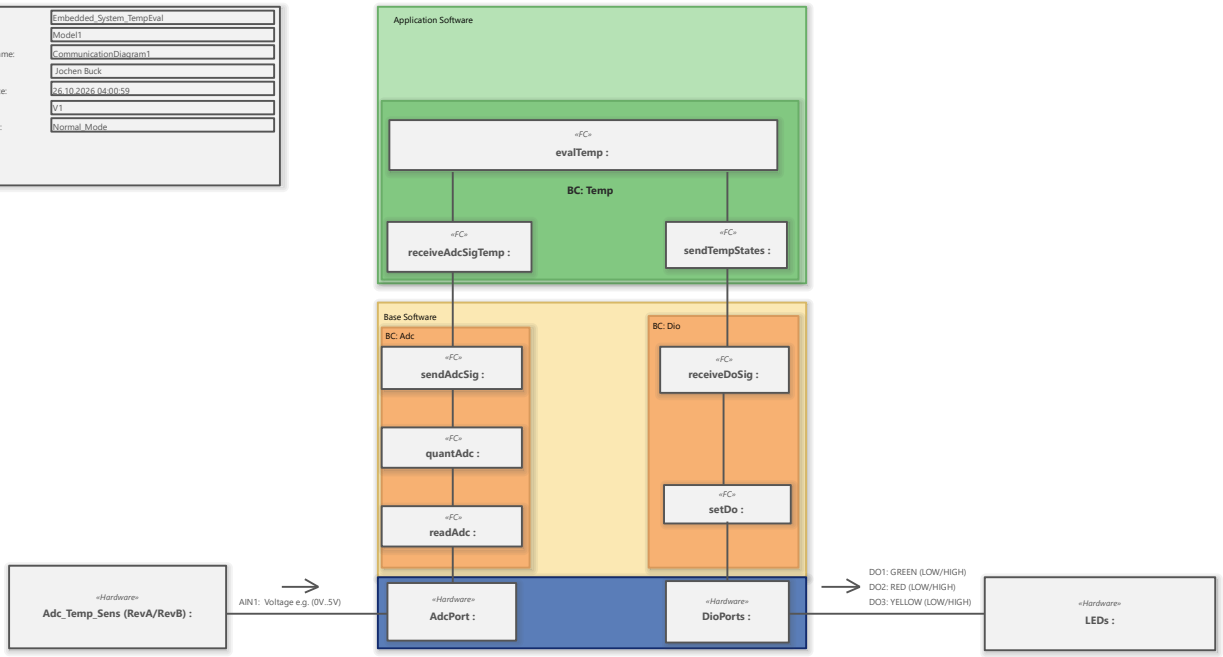
Project:	Embedded_System_TempEval
Folder:	Model1
Diagram name:	CommunicationDiagram1
Author(s):	Jochen Buck
Change date:	26.10.2026 04:00:59
Version:	V1
Description:	Normal_Mode



Explanation(general):
This is a software component communication diagram. All related software components (SWC), which are necessary for the normal/run operation are visible. The basic structure of the architecture is a separation between ASW (apllication software) and BSW (Base Software). In ASW are SWCs which are not directly related to Hardware. In Base Software are all the SWCs with Hardware dependency. The SWCs are also clustered. There are Base Components (BCs), which summerize several functionalities. These functionalities are named as Functional Components (FCs). The FCs have communicate with values or states between each other. On the buttom of the diagram there is also the Hardware layer shown, with in and outputs.

Explanation (functional): Beginning on Left:
Analog signal/Temperature Sensor is connected to AdcPort of uC. Within the AdcPort conversion and sampling is done.
FC: readAdc is geting converted and sampled analog value from AdcPort HW Registers of uC

FC: qauntAdc performs quantization depending on Adc_Temps_Sens type (voltage -> digits)

FC: sendAdcSig: is providing digits value to ASW BC:Temp

FC: receiveAdcSigTemp is receiving digit value of BSW BC:Adc and converting value in temperature value in Deg Celcius. Deg Celsius value is provided to FC:evalTemp

FC:evalTemp is evaluating temperature value as NORMAL, CRITICAL or WARNING: These states are provided to FC:sendTempStates

FC:sendTempStates: Is sending temperature states to BSW BC:Dio

FC:receiveDoSig is receiving temperature states from ASW BC:Temp and sending them further to FC:setDo

FC:setDo is receiving temperature states and is setting digital Output (Do) Registers of uC according the temperature states

Project:	Embedded_System_TempEval
Folder:	Model1
Diagram name:	CommunicationDiagram2
Author(s):	
Change date:	22.10.2026 16:24:29
Version:	
Description:	Init_Mode

